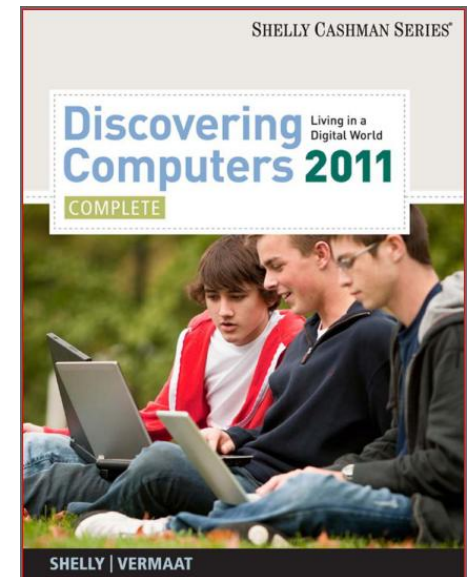


# Lecture 4:

## Instrument Types

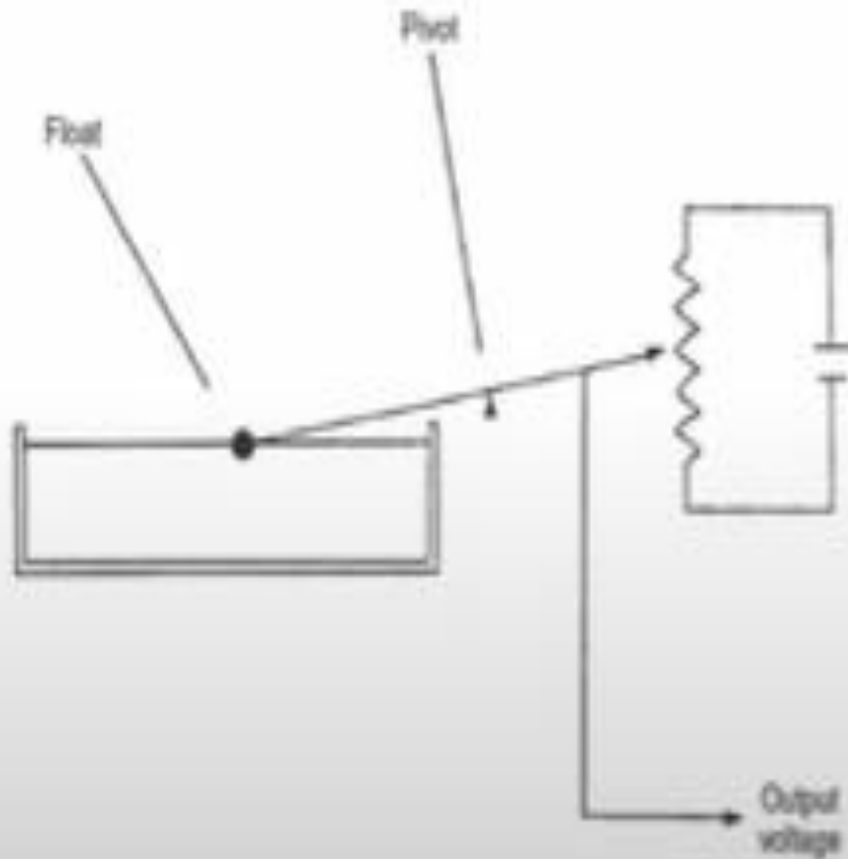


# Instrument Classification

---

- *Active and Passive Instrument.*
- *Null and Deflection Type Instruments*
- *Analogue and Digital Instruments*
- *Contact and Non-contact Type*
- *Intelligent and Dump Type*

# Active and Passive Instrument

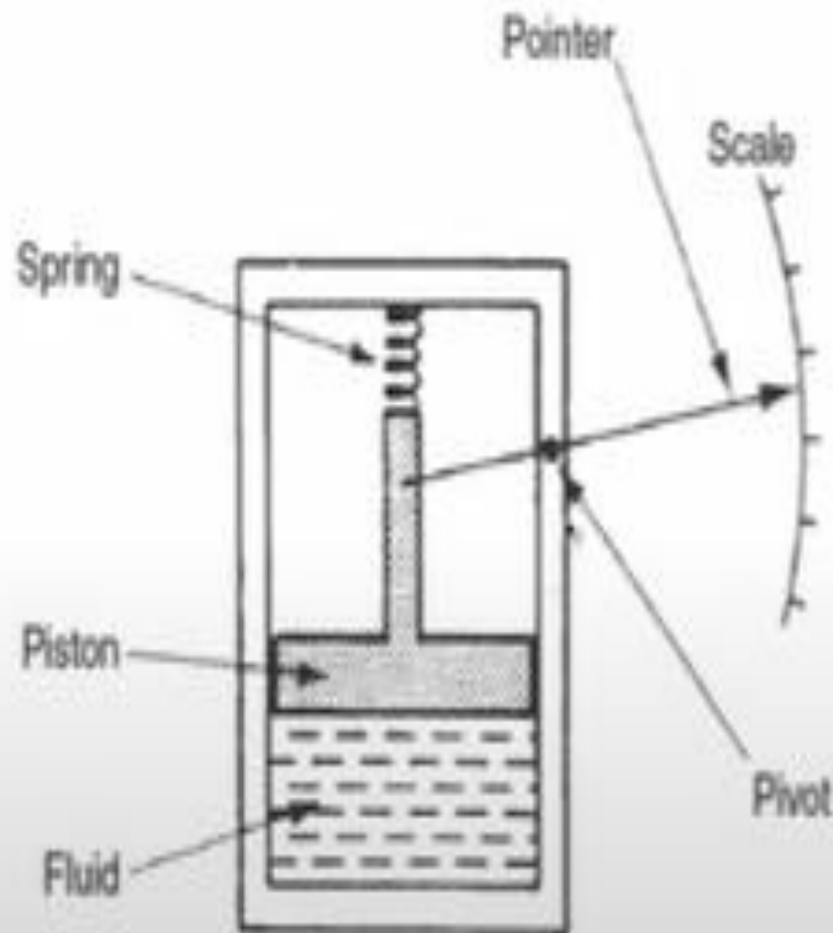


*Petrol-Tank Level Indicator*

## *Active Type:*

- *The energy in the output signal comes from the external power source*
- *Primary transducer simply modulates the value of the voltage from this external power source*
- *Resolution – Greater control over the measurement resolution*
- *Relatively expensive*

# Active and Passive Instrument

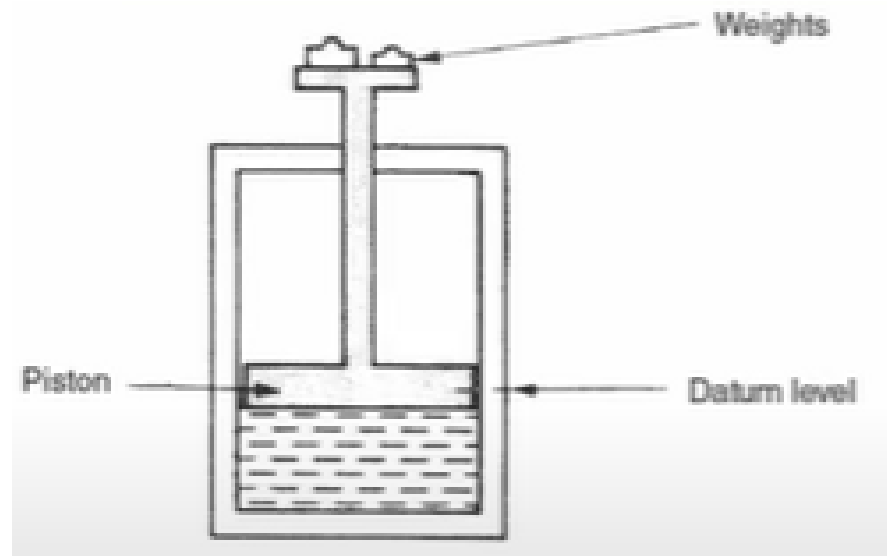


Pressure Gauge

## *Passive Type:*

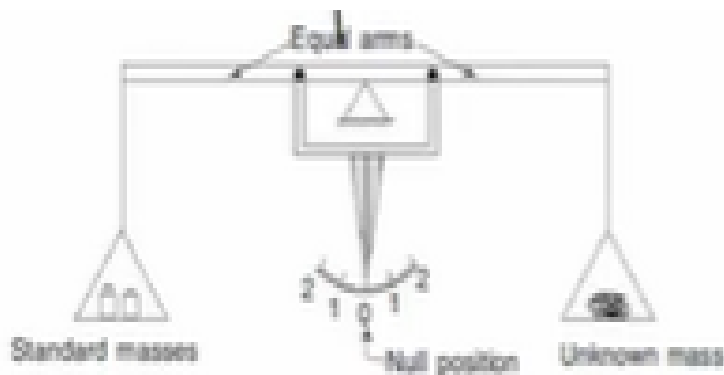
- *The output is entirely produced by the quantity being measured.*
- *No other energy inputs*
- *Resolution – Depends on practical limits of pointer arm etc.*
- *Relatively cheaper*

# Null and Deflection Type

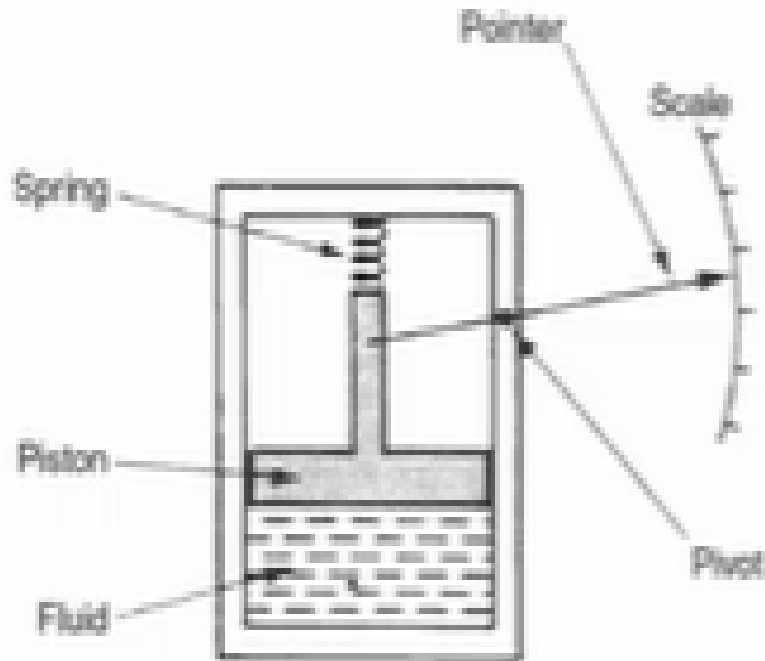


## *Null Type:*

- *Dead weight gauge*
- *Weights are put on top of piston till the downward force balances fluid pressure*
- *Weights are added till datum level is reached*
- *More accurate*



# Null and Deflection Type

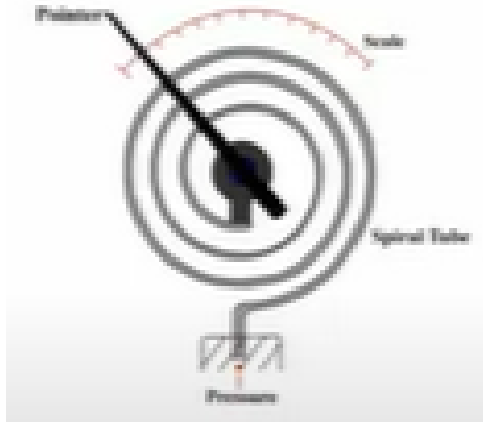


*Deflection type Pressure Gauge*

## *Deflection Type:*

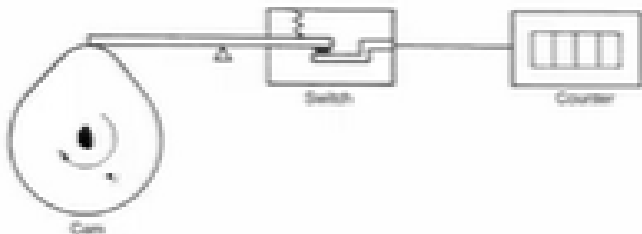
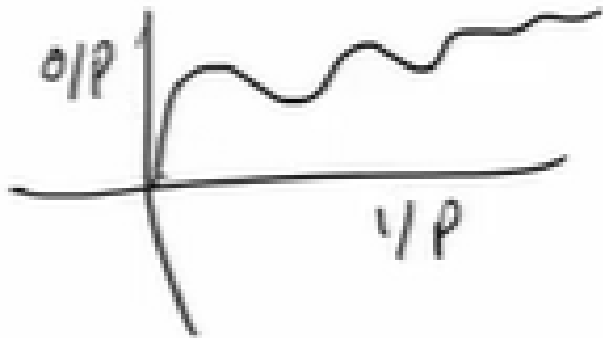
- *The value of quantity being measured is displayed in terms of the amount of movement of pointer*
- *Usage wise – Convenient comparatively – Mostly preferred in workplace*
- *Careful choice and calibration of spring is sometimes required*
- *Null type is comparatively accurate and used for calibration duties*

# Analogue and Digital Instruments

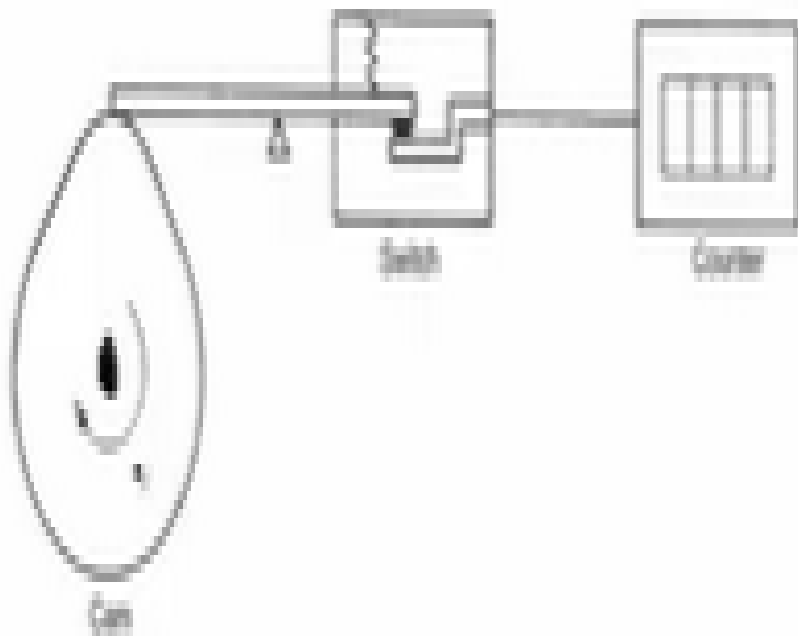


## *Analog Type:*

- *Step less variations with respect to time.*
- *These instruments usually consist of simple functional elements.*
- *Cost less and are easy to maintain and repair.*
- *Bourdon Gauge measure the magnitude of pressure in the analog form.*



# Analogue and Digital Instruments



## *Digital Type:*

- *Discrete Steps*
- *Immune to Noise*
- *Easy interface to Computer*

Disadvantage: *Cannot provide values in between two steps*



# Contact and Non-Contact Type

- Contact type:

- *Kept in the measuring medium itself*
- *Usually modifies the measurand*
- *Clinical thermometer*



Bourdon Pressure gauge  
*Cannot provide values in  
between two steps*

From the environment to the embedded systems,  
from the embedded systems to the applications

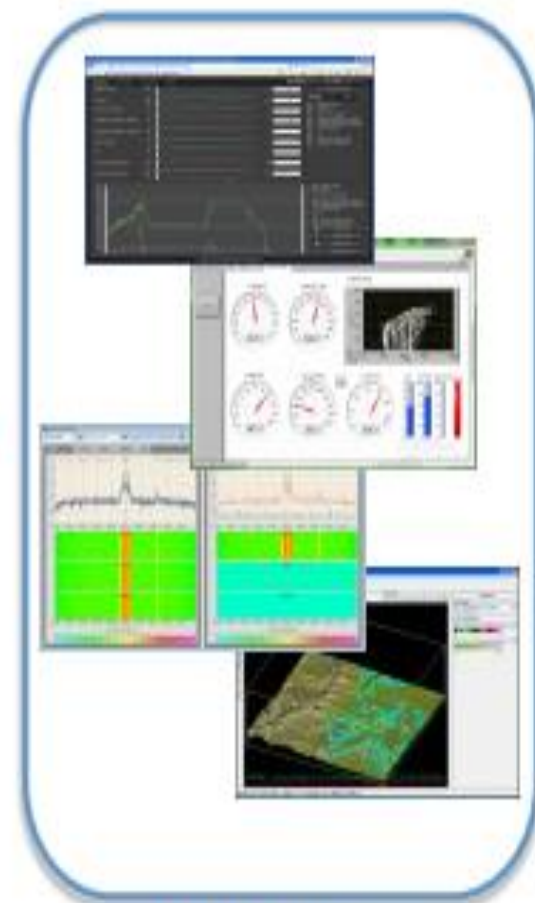
- The **emergence of non-trivial embedded sensor units, networked embedded systems and sensor/actuator networks** has made possible the design and implementation of several sophisticated applications.
- Main characteristic: **large amounts of real-time data** are collected to constitute a big data picture as time passes.
- Acquired data are then processed at local, cluster-of-units or server level to **take the appropriate actions or make the most appropriate decision**



Environment



Embedded,  
Cyber-physical  
or Pervasive  
System



Applications

## Changes

(Nonstationary,  
seasonality,  
periodicity)



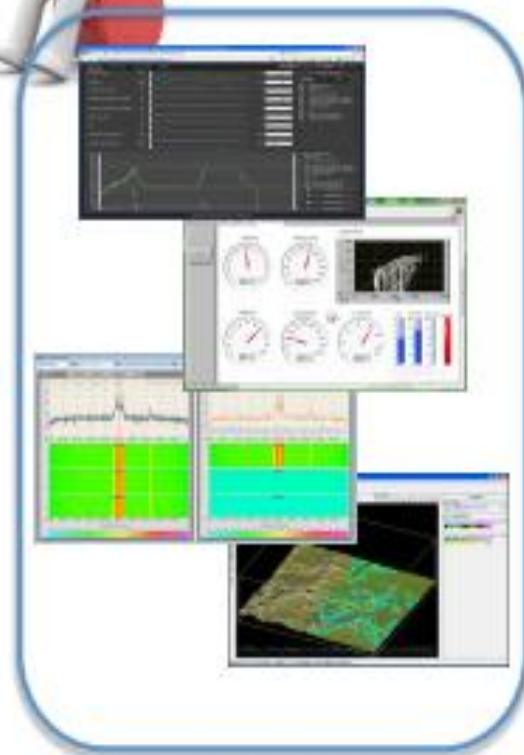
Environment

Faults, Uncertainty,  
Errors, memory  
energy constraints,  
finite precision



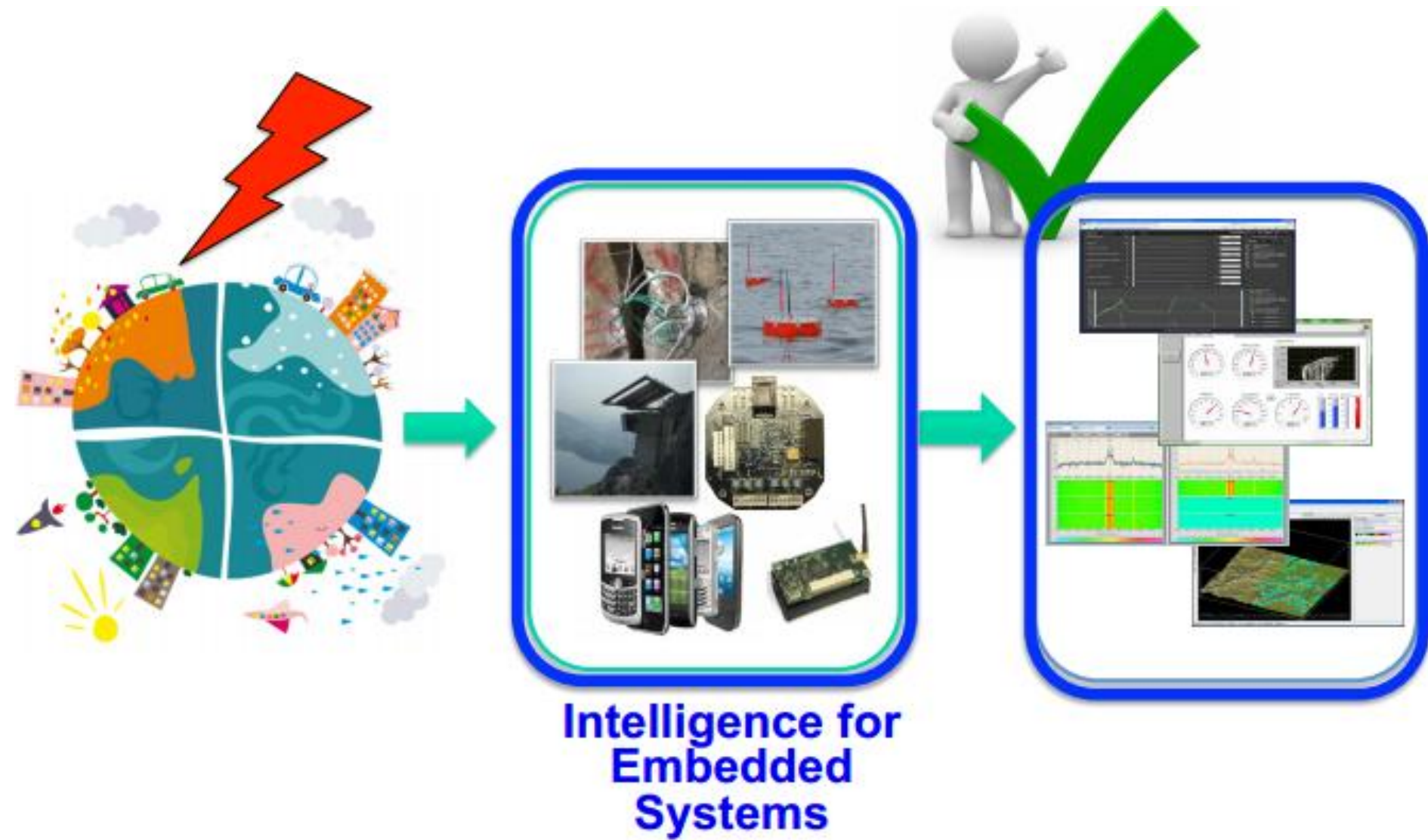
Embedded,  
Cyber-physical  
or Pervasive  
System

Performance  
reduction,  
cascade  
effects



Applications



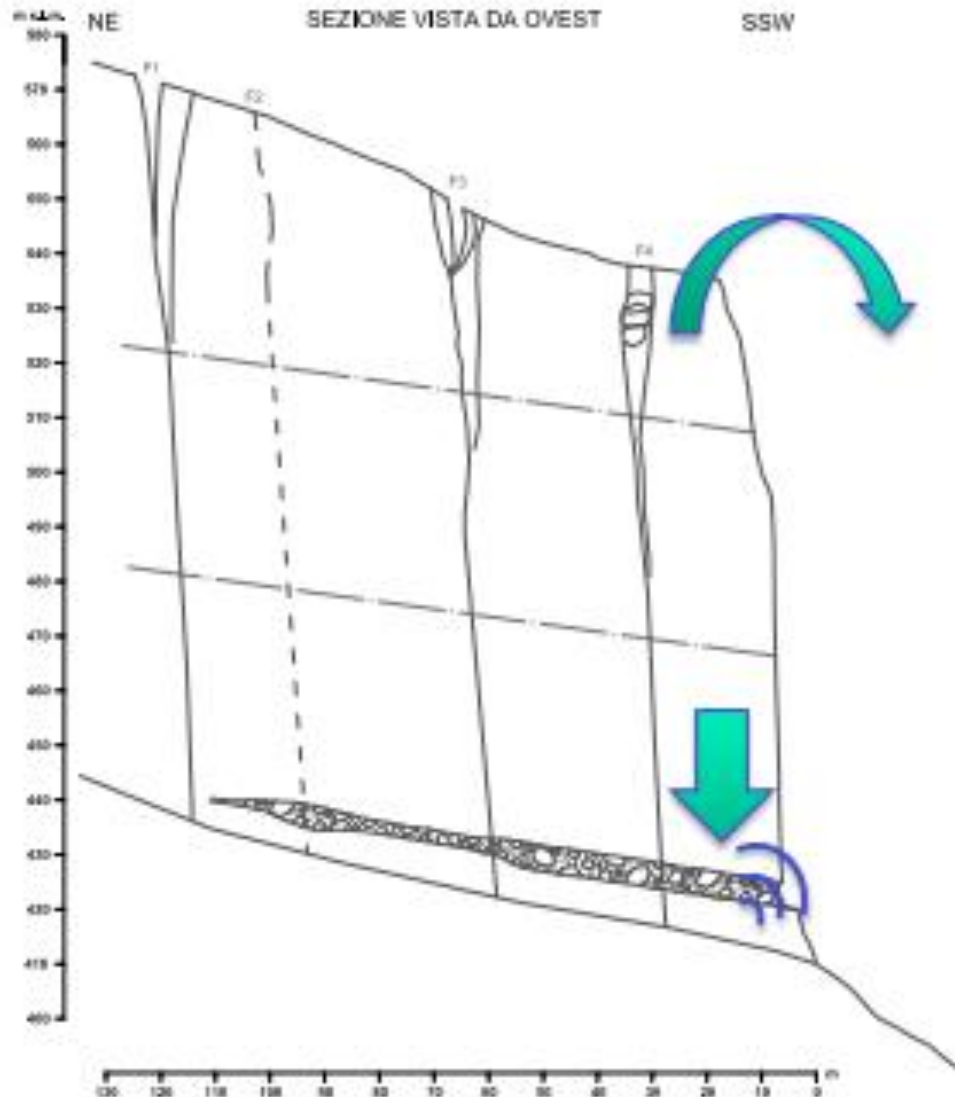


# Measure and Measurements System

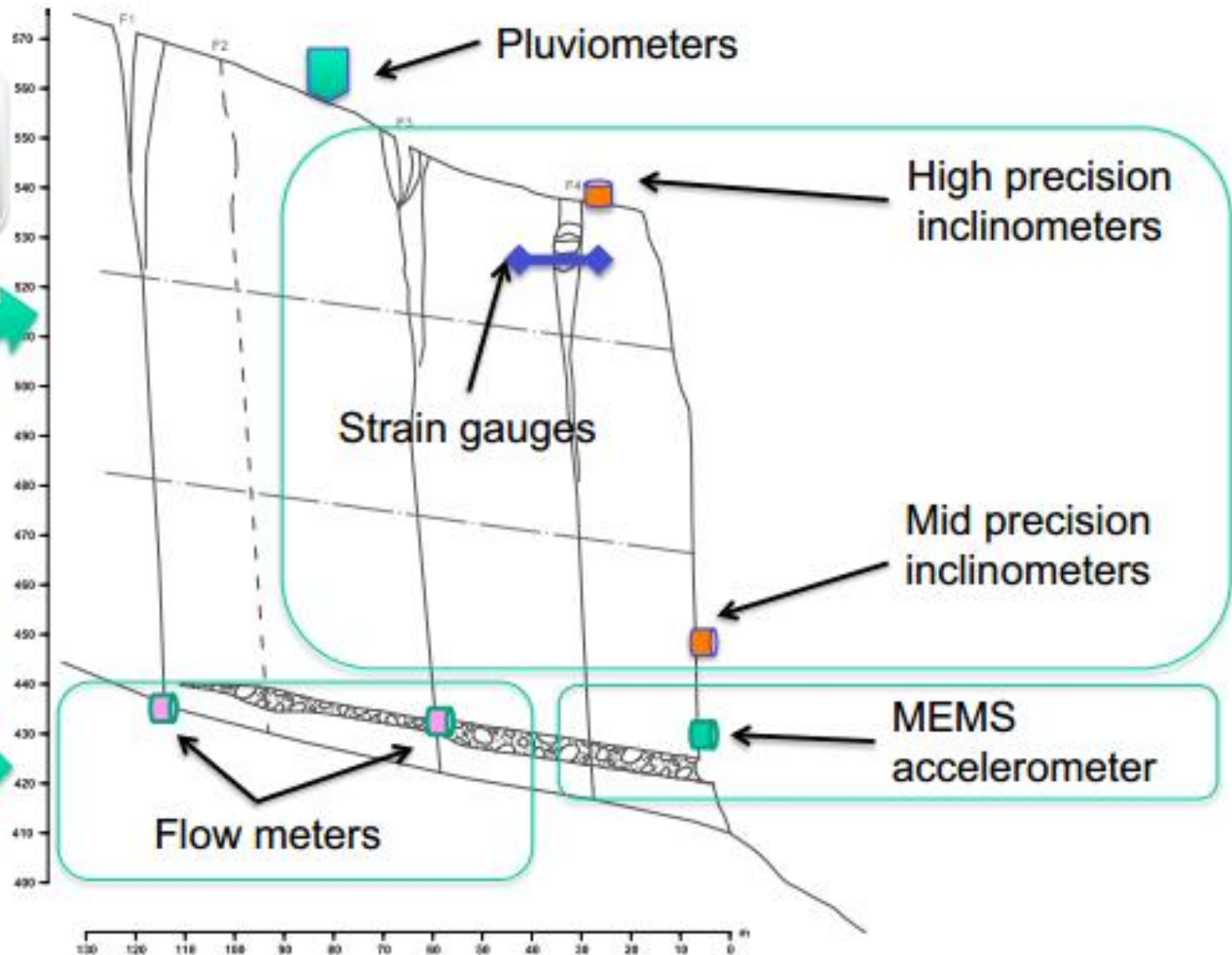
- The operation of **measuring an unknown quantity  $x_0$**  can be **modeled as taking** an instance -or measurement-  **$x_i$  at time  $i$**  with an **ad-hoc sensor  $S$**
- Even though  $S$  has been suitably designed and realized, **the physical elements** composing it **are far from being ideal** and **introduce sources of uncertainty** in the measurement process
- As a consequence,  **$x_i$  represents only an estimate of  $x_0$**

This is a crucial point for the application  
processing the data

# Environmental monitoring: Rock collapse forecasting



**In addition:**  
Many temperature  
sensors









Is  $x_i$  an accurate and reliable estimate of  $x_0$  ?



Despite the formalization of the measurement process, **several major aspects need to be investigated and addressed** before claiming that a generic measurement  $x_i$  is an accurate and reliable estimate of  $x_0$ .

■ **For instance:**

- the sensor could be noisy;
- the sensor or the processing electronics could be faulty;
- the temperature could affect the sensor inducing bias in the measurements;

How can we say that  $x_i$  is an accurate and reliable estimate of  $x_0$  ?

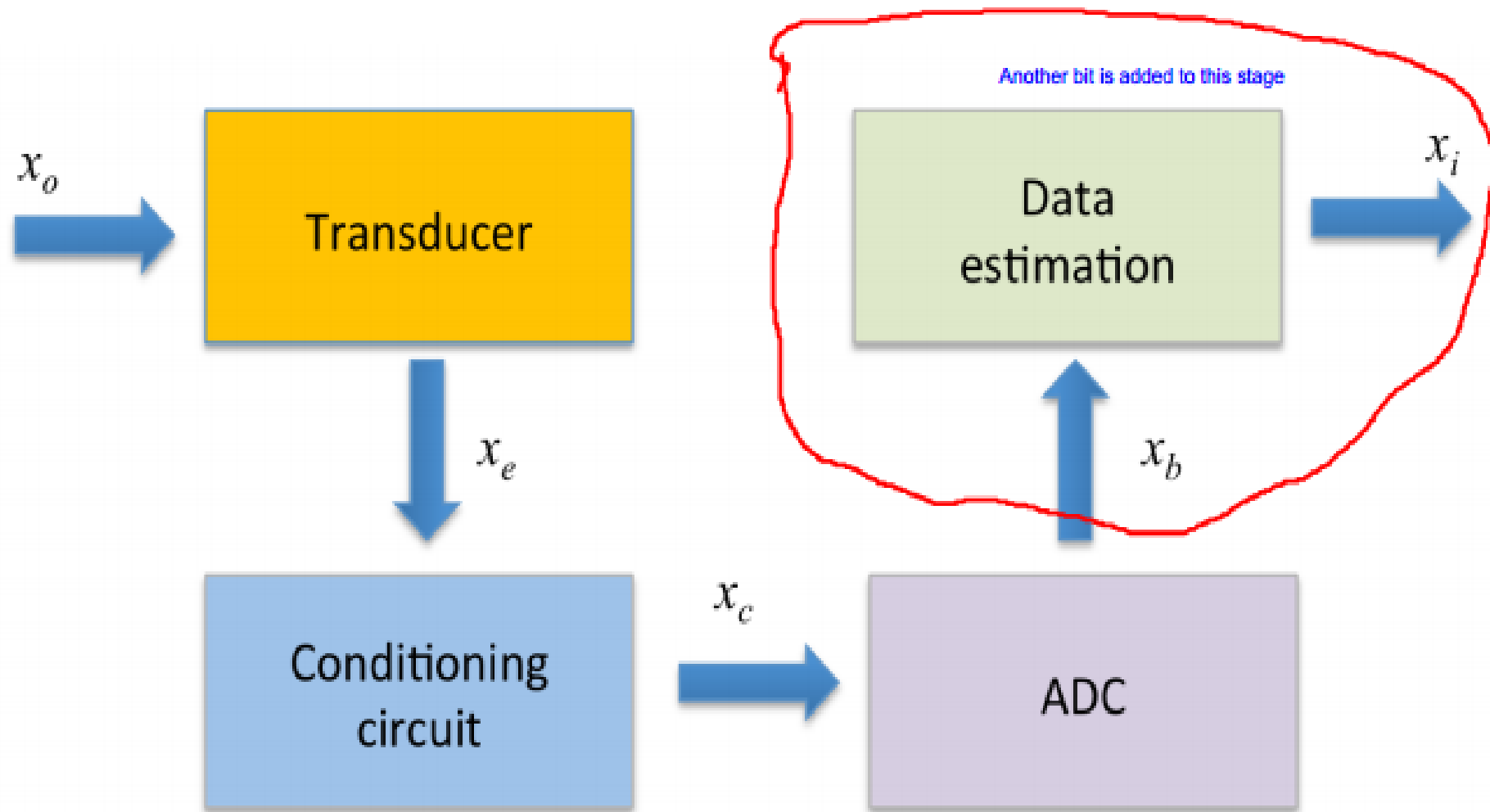


## Some properties...

- We would require **subsequent measurements  $x_i$  to be somehow centered around  $x_0$** : an accurate sensor does not introduce some bias error (*accuracy* property).
- Each measurement represents only an estimate of the true unknown value, **the discrepancy** between the two - or error- **depends on the quality of the sensor and the working conditions** under which the measure was taken (*precision* property).
- We hope that the sensor is able to **provide a long sequence of correct digits** of the number associated with the acquired data (*resolution* property).



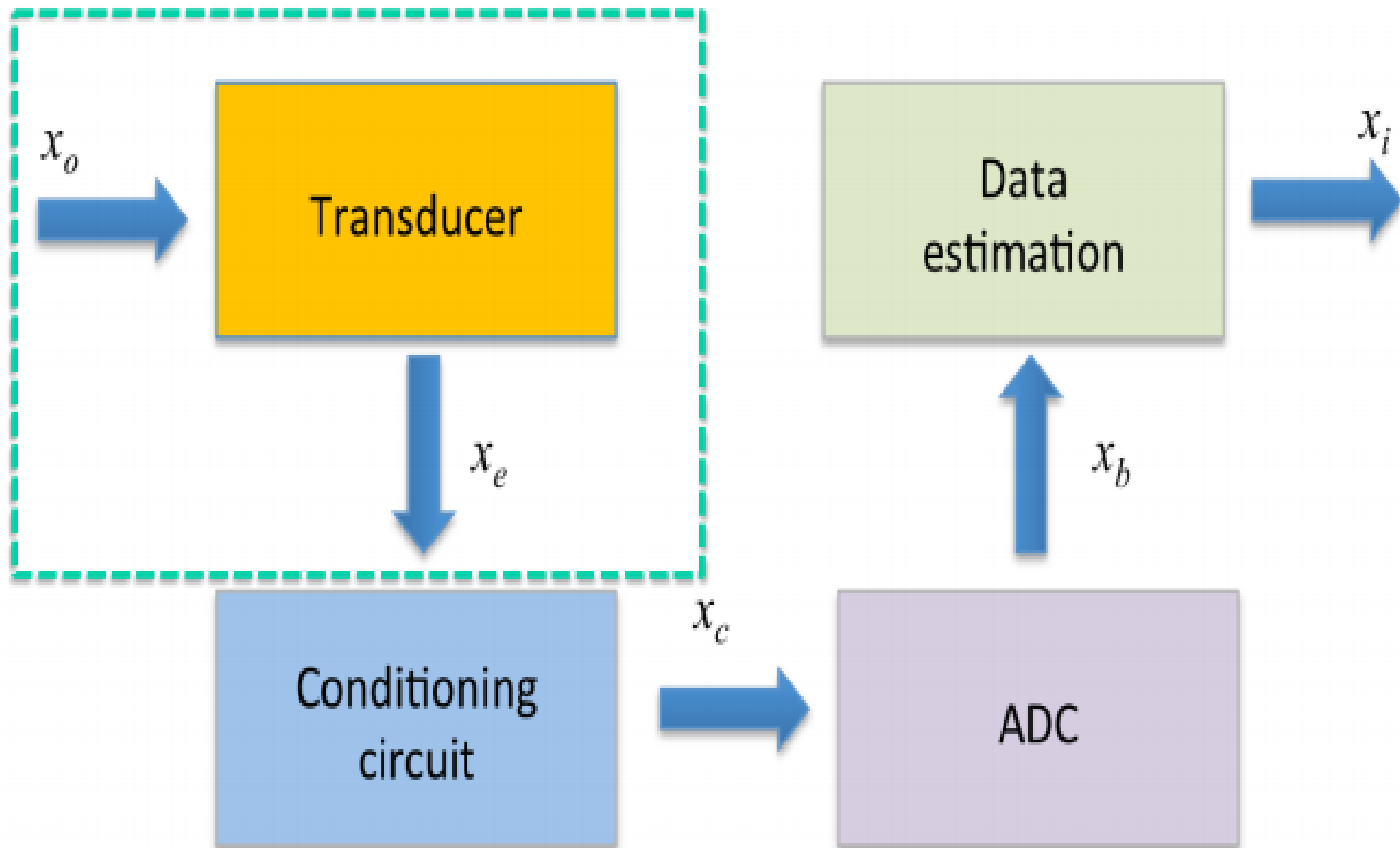
## The measurement chain



*The functional chain representing the most common model describing a modern electronic sensor*



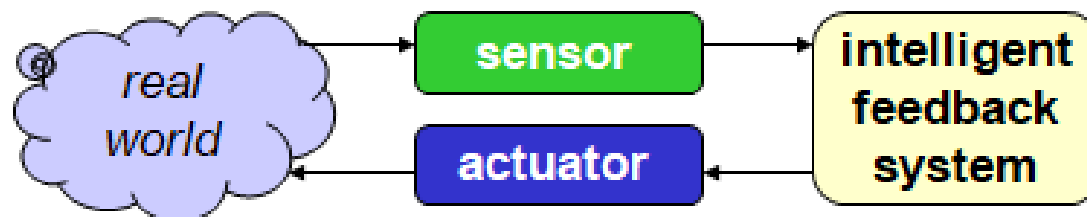
## The transducer: from $x_0$ to $x_e$



# Transducers

---

- **Transducer**
  - a device that converts a primary form of energy into a corresponding signal with a different energy form
    - Primary Energy Forms: mechanical, thermal, electromagnetic, optical, chemical, etc.
  - take form of a **sensor** or an **actuator**
- **Sensor** (e.g., thermometer)
  - a device that detects/measures a signal or stimulus
  - acquires information from the "real world"
- **Actuator** (e.g., heater)
  - a device that generates a signal or stimulus





# CLASSIFICATION OF TRANSDUCERS

On the basis of transduction form used.

As primary and secondary transducers.

As passive and active transducers.

As analog and digital transducers.

As transducers and inverse transducers

# SELECTION CRITERIA OF THE TRANSDUCERS

Operating principle

Sensitivity

Operating range

Accuracy

Errors

Environmental capability

Insensitive to unwanted Signal

Stability

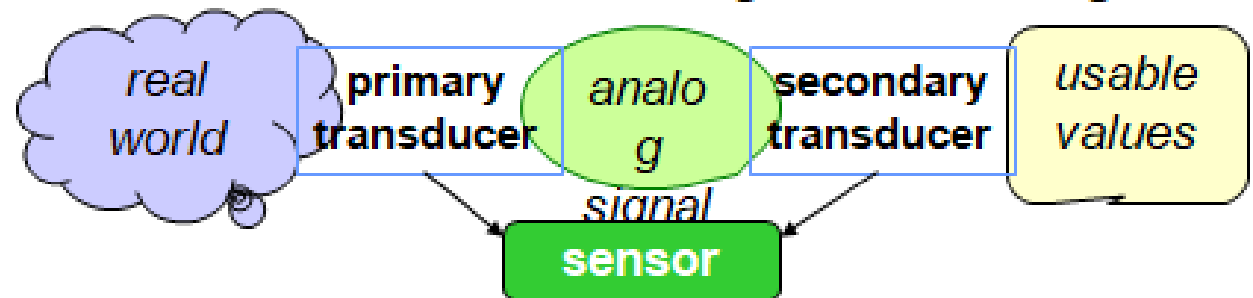
# Sensor Systems

Typically interested in **electronic sensor**

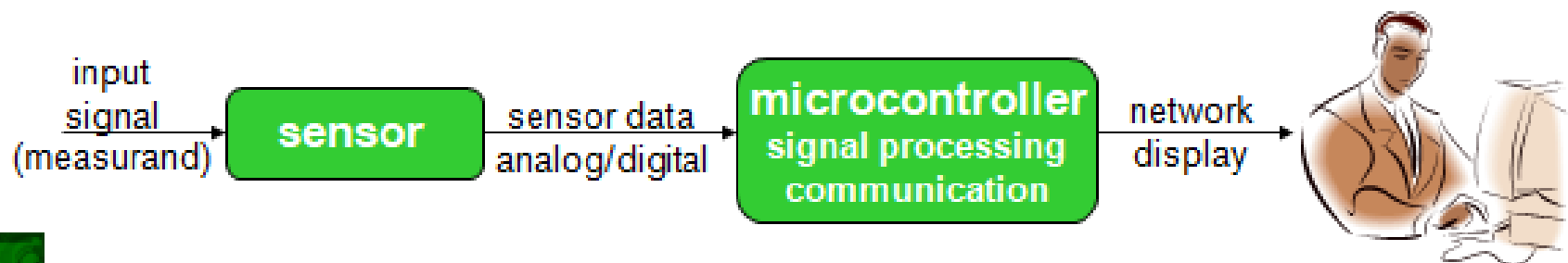
- convert desired parameter into electrically measurable signal

- **General Electronic Sensor**

- primary transducer: changes "real world" parameter into electrical signal
- secondary transducer: converts electrical signal into analog or digital values



- **Typical Electronic Sensor System**



# Example Electronic Sensor Systems

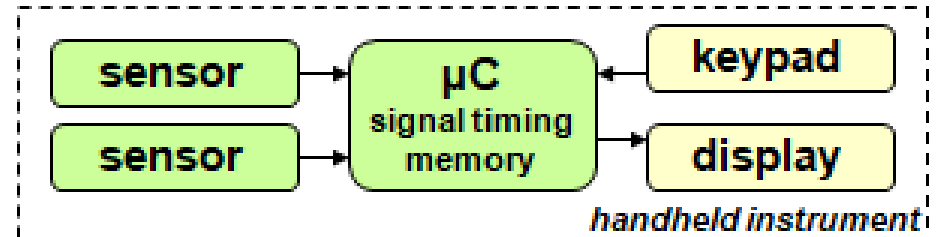
- Components vary with application

- digital sensor within an instrument

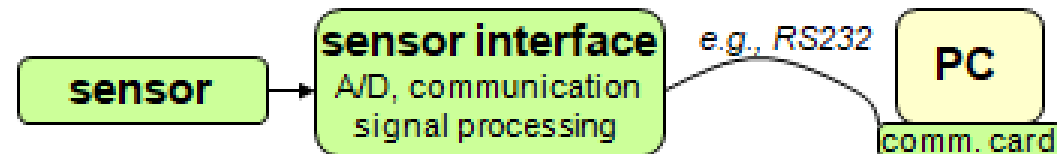
- microcontroller

- signal timing

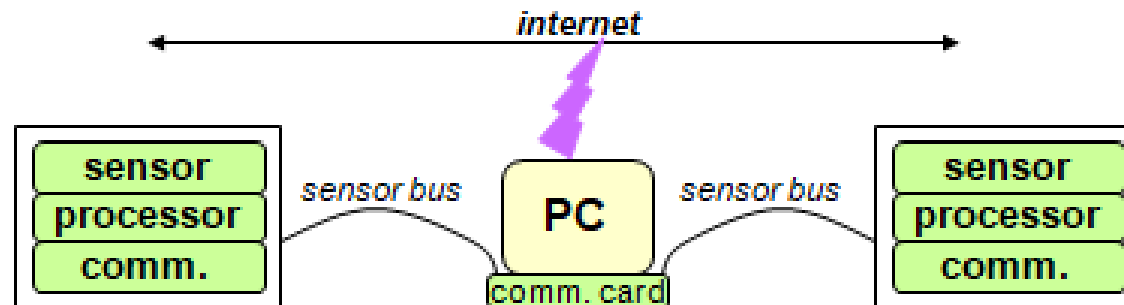
- data storage



- analog sensor analyzed by a PC



- multiple sensors displayed over internet



# Primary Transducers

---

- Conventional Transducers

*large, but generally reliable, based on older technology*

- thermocouple: temperature difference
- compass (magnetic): direction

- Microelectronic Sensors

*millimeter sized, highly sensitive, less robust*

- photodiode/phototransistor: photon energy (light)
  - infrared detectors, proximity/intrusion alarms
- piezoresistive pressure sensor: air/fluid pressure
- microaccelerometers: vibration,  $\Delta$ -velocity (car crash)
- chemical sensors:  $O_2$ ,  $CO_2$ , Cl, Nitrates (explosives)
- DNA arrays: match DNA sequences



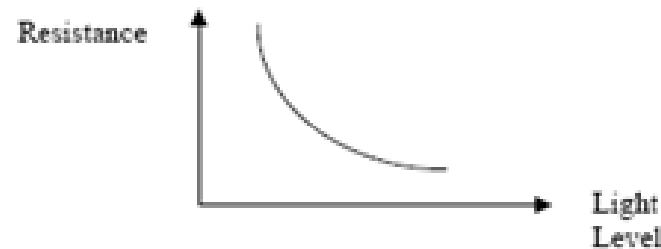
# Example Primary Transducers

- Light Sensor

- photoconductor

- light  $\rightarrow \Delta R$

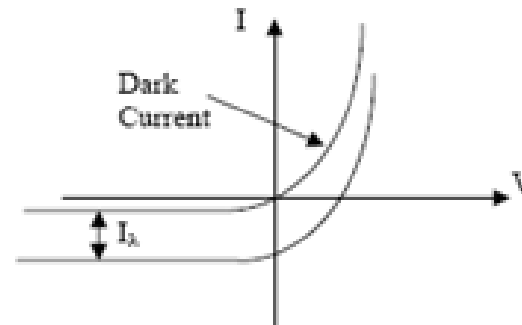
Photoconductor: (Light sensitive semiconductor resistor)



- photodiode

- light  $\rightarrow \Delta I$

Photodiode:

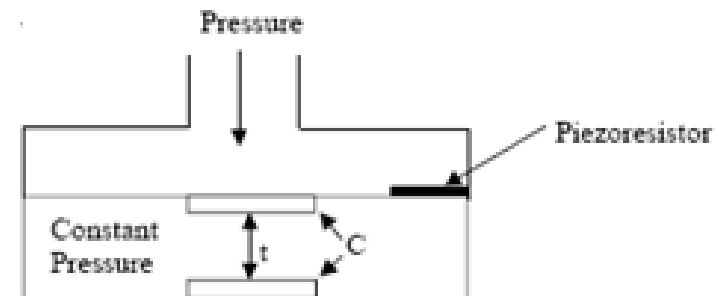


$$I = I_0 [\exp(eV/kT) - 1] - I_\lambda$$

$I_\lambda$  is proportion to the light level

- membrane pressure sensor

- resistive (pressure  $\rightarrow \Delta R$ )
- capacitive (pressure  $\rightarrow \Delta C$ )



$$C = \epsilon A / t$$



# Displacement Measurements

---

- Measurements of size, shape, and position utilize **displacement sensors**
- Examples
  - diameter of part under stress (direct)
  - movement of a microphone diaphragm to quantify liquid movement through the heart (indirect)
- Primary Transducer Types
  - Resistive Sensors (Potentiometers & Strain Gages)
  - Inductive Sensors
  - Capacitive Sensors
  - Piezoelectric Sensors
- Secondary Transducers
  - Wheatstone Bridge
  - Amplifiers

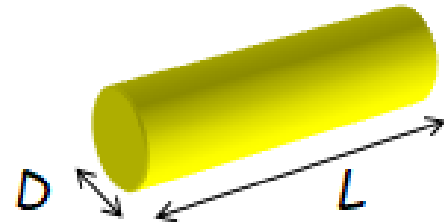




# Strain Gage: Gage Factor

- Remember: for a strained thin wire

- $\Delta R/R = \Delta L/L - \Delta A/A + \Delta \rho/\rho$ 
    - $A = \pi (D/2)^2$ , for circular wire



- Poisson's ratio,  $\mu$ : relates change in diameter  $D$  to change in length  $L$

- $\Delta D/D = -\mu \Delta L/L$

- Thus

- $\Delta R/R = (1+2\mu) \Delta L/L + \Delta \rho/\rho$

dimensional effect    piezoresistive effect

- Gage Factor,  $G$ , used to compare strain-gate materials

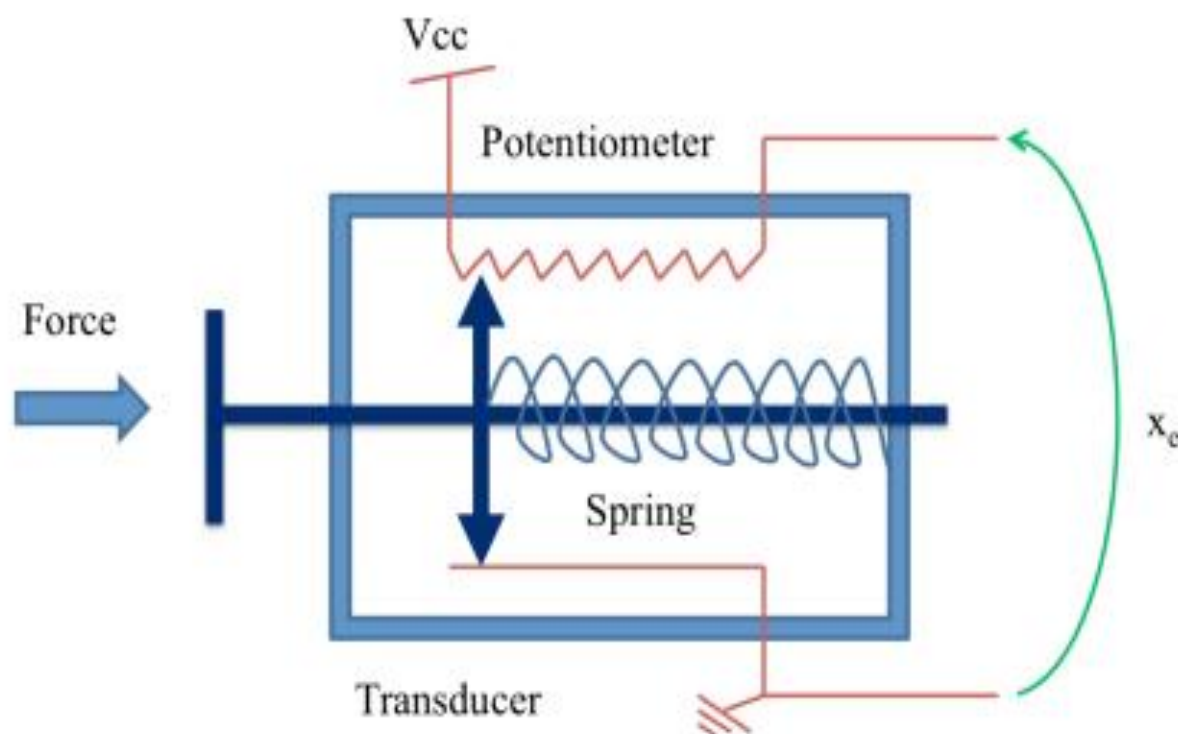
- $G = \frac{\Delta R/R}{\Delta L/L} = (1+2\mu) + \frac{\Delta \rho/\rho}{\Delta L/L}$





## The transducer

- A transducer is a **device transforming a form of energy into another**, here converting a physical quantity  $x_o$  into an electric or electric-related quantity  $x_e$
- *E.g., a force transducer*





## Transducer (2)

- **Sensors can be active or passive in their transduction mechanism:** an active sensor requires energy to carry out the operation and needs to be powered, whereas a passive sensor does not

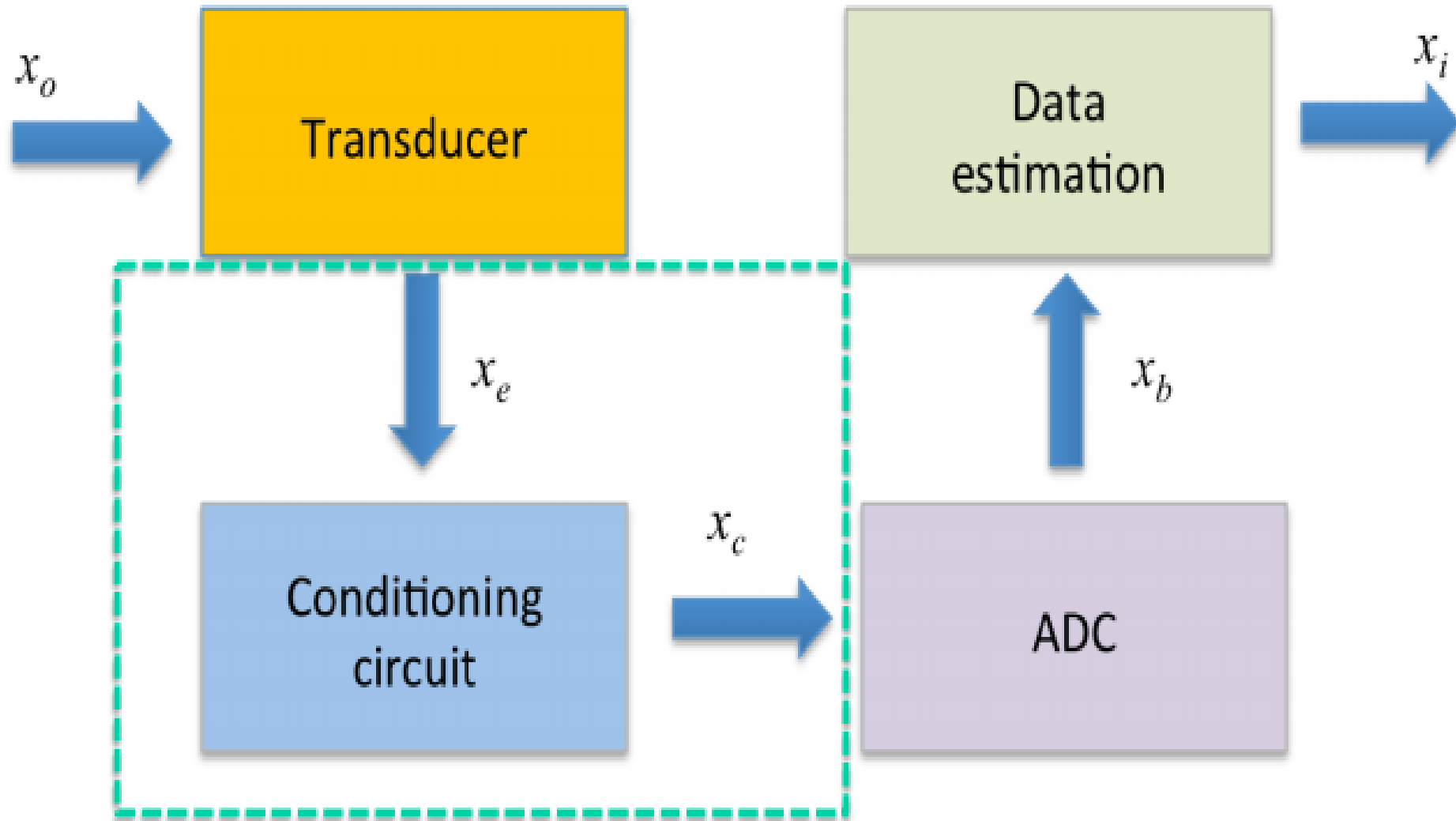
➡ *Required Energy to measure  $x_e$*

- **Another relevant information is related to the time requested to produce a stable measurement.** Such a time depends, for instance, on the dynamics of the transduction mechanisms or the time needed to complete the self calibration/compensation phase introduced to improve the quality of the sensor outcome.

➡ *Required Time to measure  $x_e$*



## The conditioning circuit: from $x_e$ to $x_c$





## The conditioning circuit

- The aim of the conditioning circuit is to provide an enhanced electrical quantity  $x_c$  of  $x_e$ .
- Why do I need a conditioning circuit?
  - ✓ the sensitivity of the sensor is amplified,
  - ✓ the effect of the noise mitigated,
  - ✓ the interval of definition of the electrical entity adapted to the requirements of the subsequent analog to digital converter.



## The conditioning circuit (2)

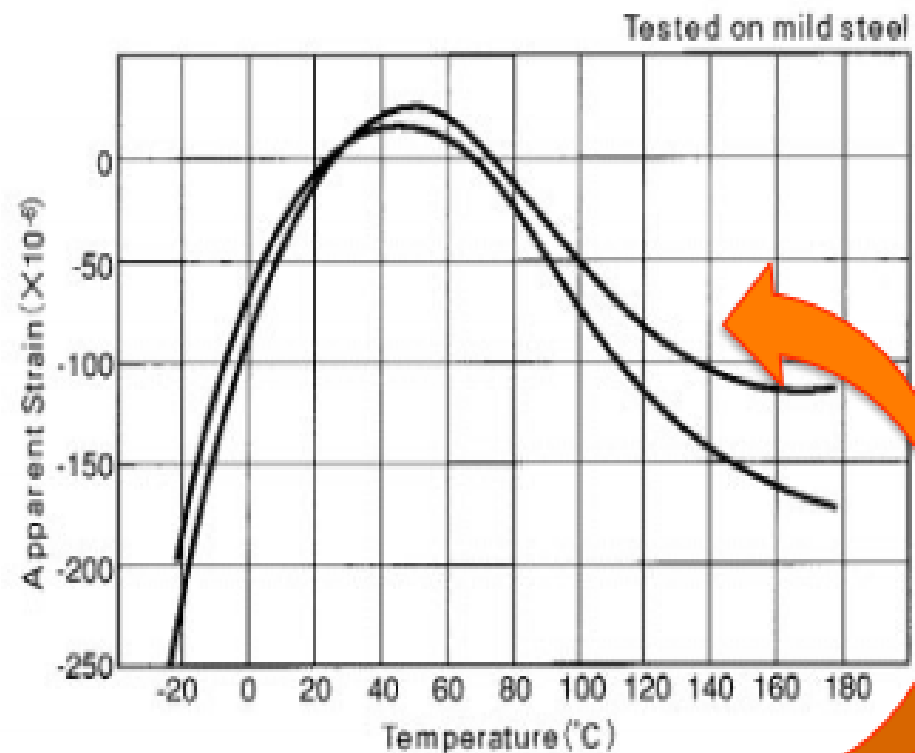
- What is the conditioning circuit?
  - It is an analog circuit juxtaposed to the transducer module
- How does it work?
  - at first **it usually amplifies  $x_e$**
  - then, **it filters its output** (e.g., with a low pass filter) **to improve the signal to noise ratio** and the quality of the signal  $x_c$  to be passed to the analog to digital conversion stage.



## The conditioning circuit (3)

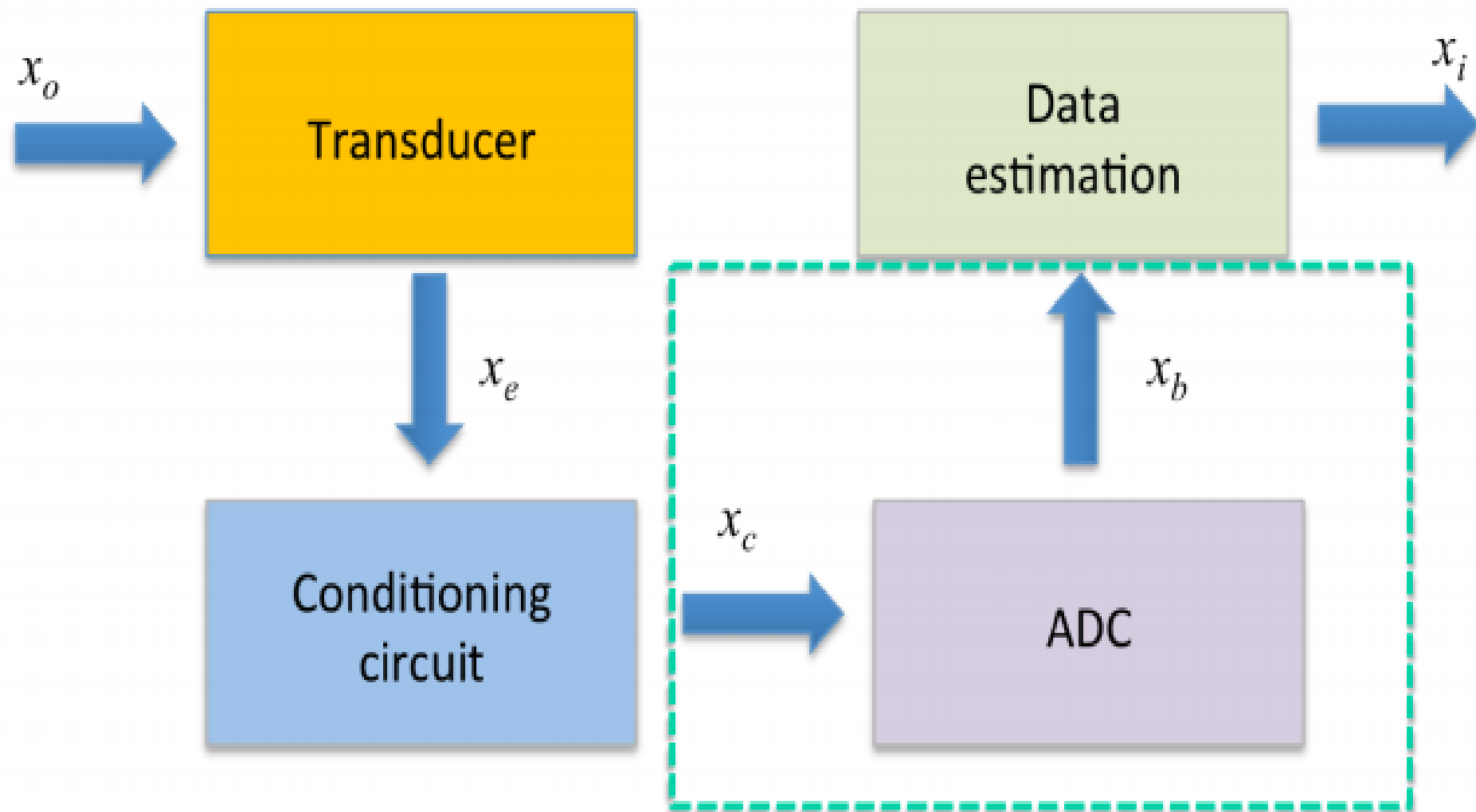
- The conditioning circuit might also encompass a module designed to:
  - *help compensating parasitic thermal effects, which influence the readout value*
  - *introduce corrections to linearize the relationship between the input  $x_o$  and  $x_c$ .*

Fig.1 Thermal Output Characteristics



Effect of the temperature on a strain gauge

## ADC: from $x_c$ to $x_b$ (binary)







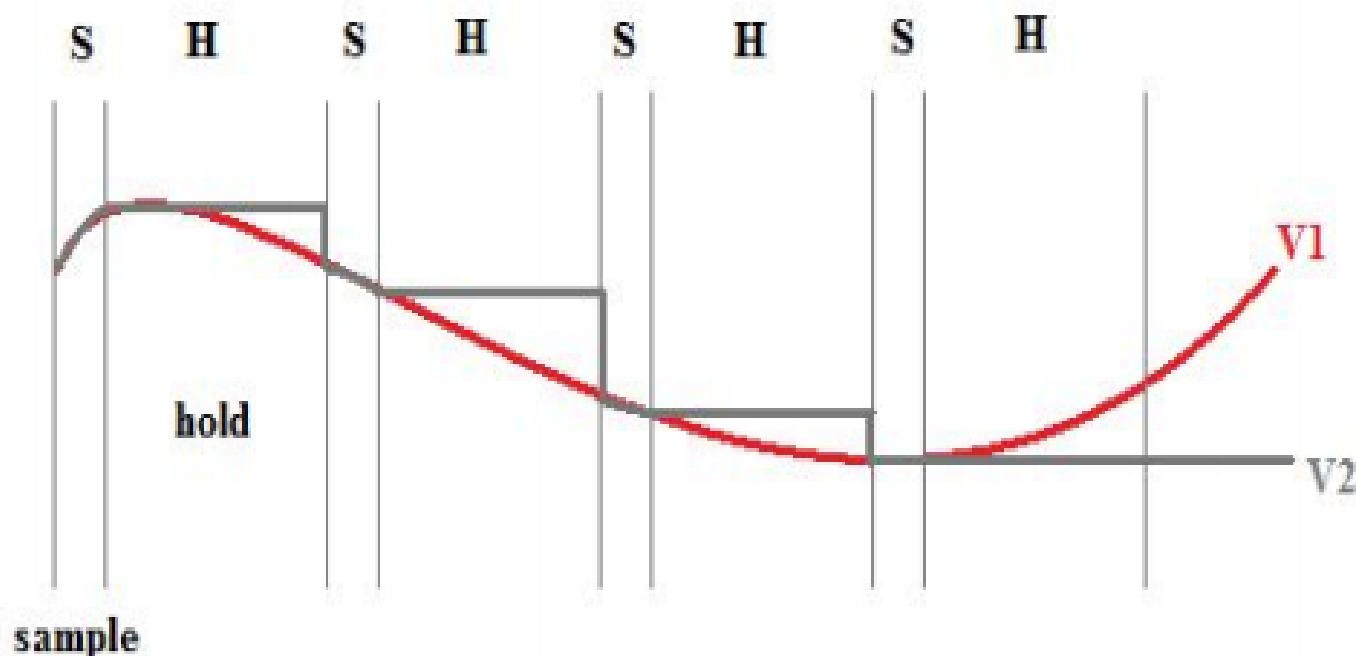
## The analog to digital converter

- The input to the module is the analog electrical signal  $x_c$  and the output is a codeword  $x_b$  represented in a binary format.
- There is a large variety of architectures for ADCs, all of them having in common:
  - the sampling rate
  - the resolution (the number of bits of the codeword)
- The conversion introduces an error associated with the quantization level, whose statistical properties may depend on the specific ADC architecture.



## The analog to digital converter (2)

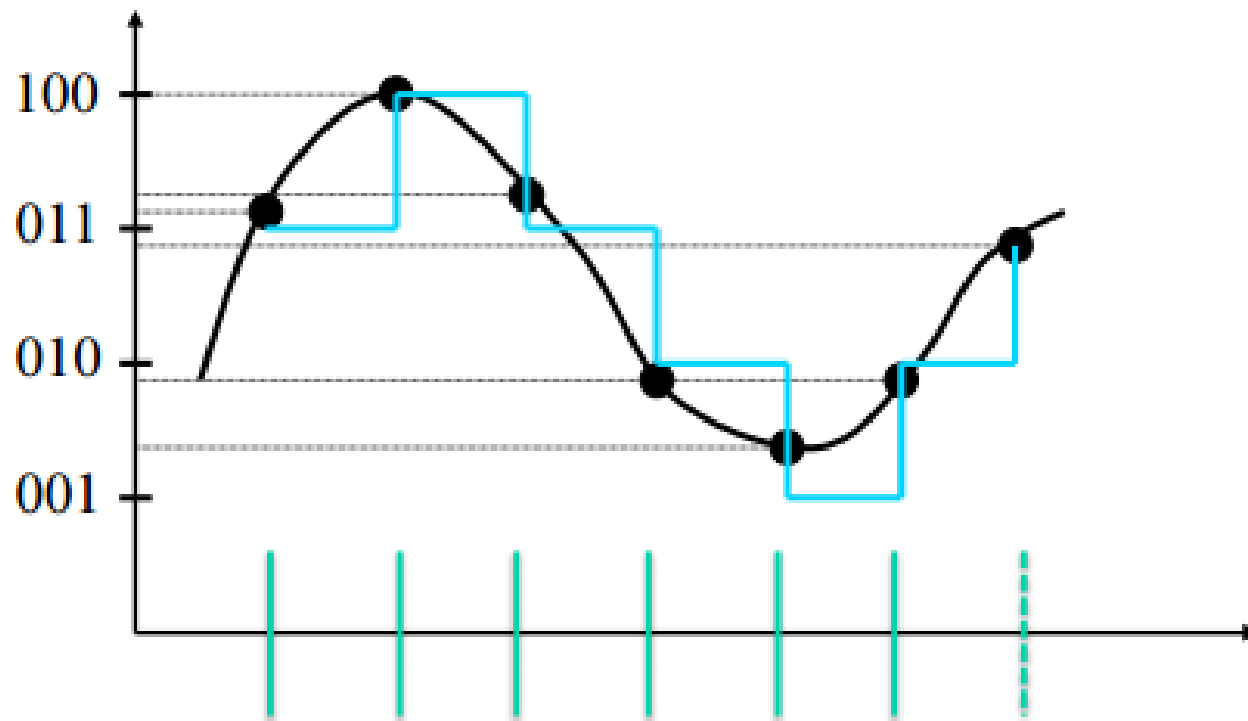
- During the conversion phase, the input  $x_c$  must be kept constant, operation carried out by the "**sample and hold**" mechanism (the analog value is sampled and kept to avoid dangerous fluctuations in the input signal)





# Sampling rates and resolution

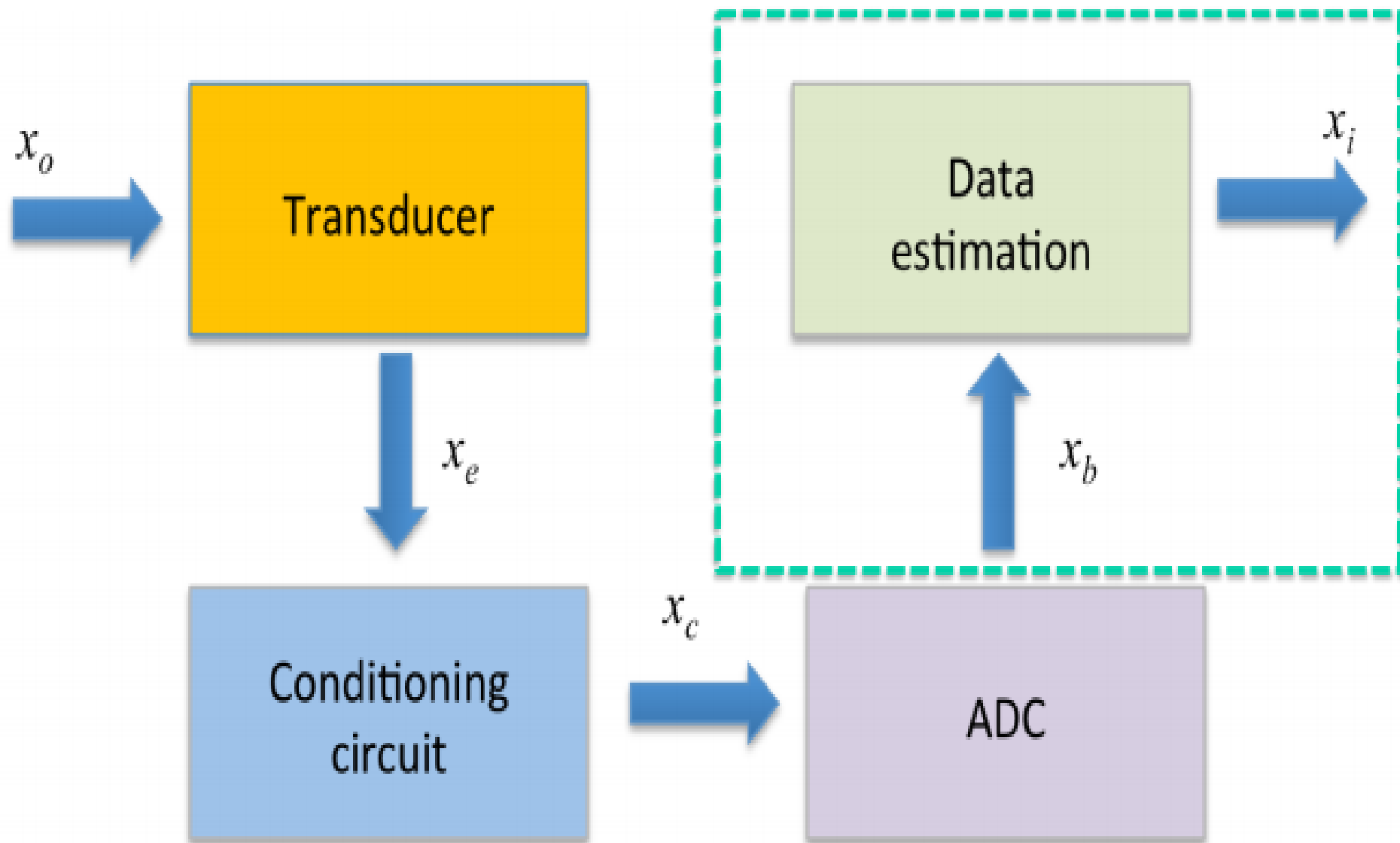
**Resolution**  
(3 bits per sample)



**Sampling Period**

(Sampling Frequency, Spectrum Analysis,  
Maximum Frequency, Nyquist theorem)

## Data Estimation: from $x_b$ to $x_i$



# The Data Estimation Module (for Smart Sensors)

- The final module (when present) **introduces further corrections on  $x_b$  by operating at the digital level.**
- In particular, it generally carries out a **further calibration phase aiming at improving the quality of the final data  $x_i$**
- **When a microprocessor is present** to address the data estimation module needs, the sensor is defined to be a **"smart sensor"**.
- The microprocessor can carry out a **more sophisticated processing** relying on simple but effective algorithms, generally aimed at **introducing corrections and structural error compensations.**

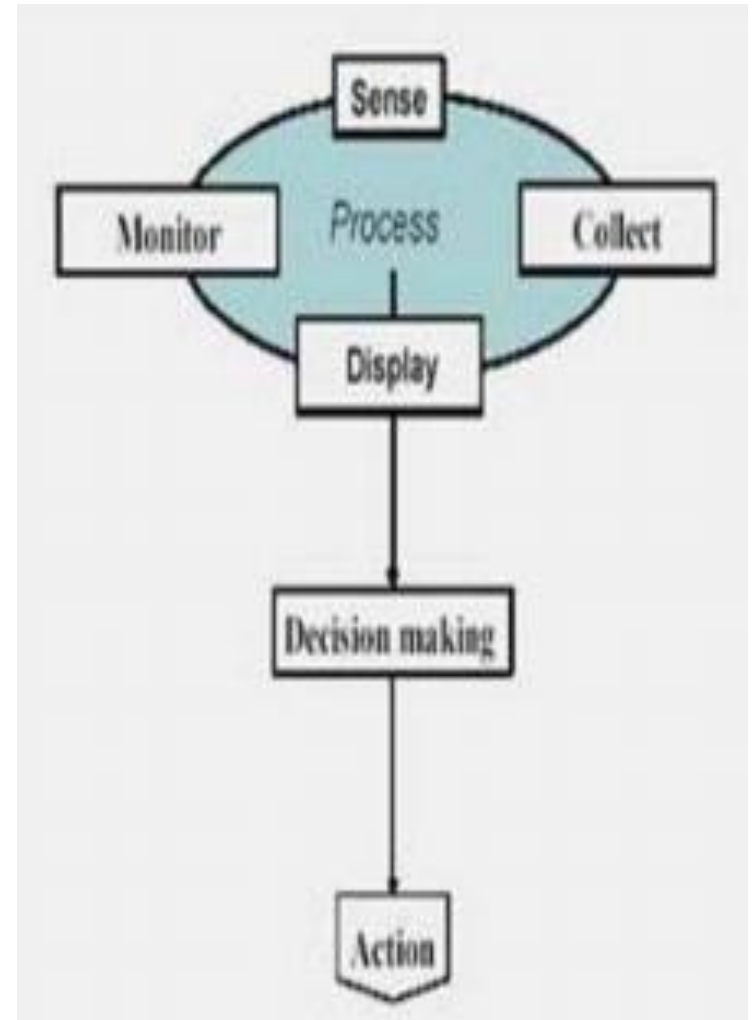
# Smart sensor

*Sensors + Interfacing hardwares = Smart sensors*

- A sensor producing an electrical output when combined with interface electronic circuits is said to be an **intelligent sensor** if the interfacing circuits can perform
  - a) Ranging
  - b) Calibration, and
  - c) Decision making for communication and utilization of data.

# Smart Sensor

- Smart sensors/  
Intelligent sensors with  
integrated electronics  
that can perform one  
or more of the following  
function:
  1. Data Conversion
  2. Bidirectional  
communication
  3. Take decisions
  4. Perform logic  
operations



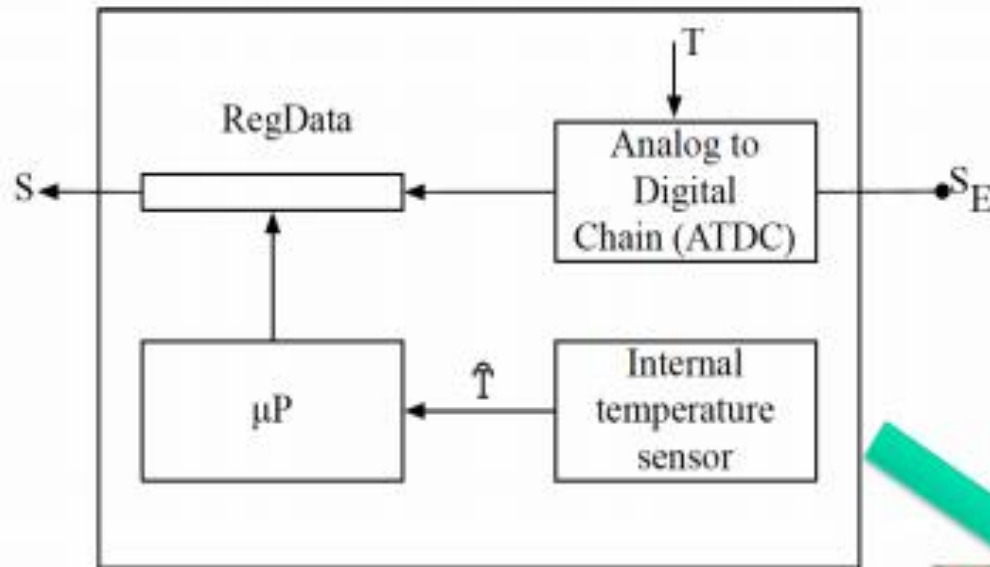
# Properties of Smart Sensor (An Intelligent field device)

- Automatic ranging and calibration
- Auto-acquisition and storage of calibration constant
- Self tuning control algorithms
- Dynamic reconfiguration should be performed
- Conditional monitoring is also used for fault diagnosis which in turn may involve additional sensors, DSP and data analysis software
- Communication through serial bus.
- Auto correction of offsets, time and temperature drifts



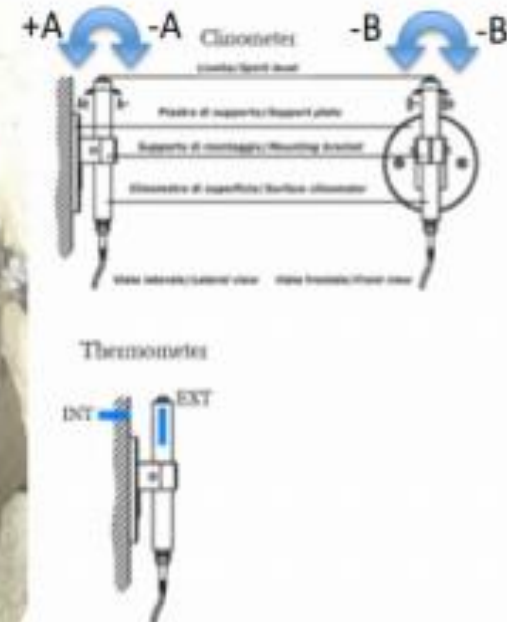


# Data Estimation Module: an example of smart clinometer sensor



For instance, a **thermal sensor can be onboard, in addition to the principal sensor**, to compensate the thermal effect on the principal sensor readout.

The microprocessor carries out the thermal compensation by reading the temperature value, comparing it with the rated working temperature defined at design time and introducing a correction on the readout value





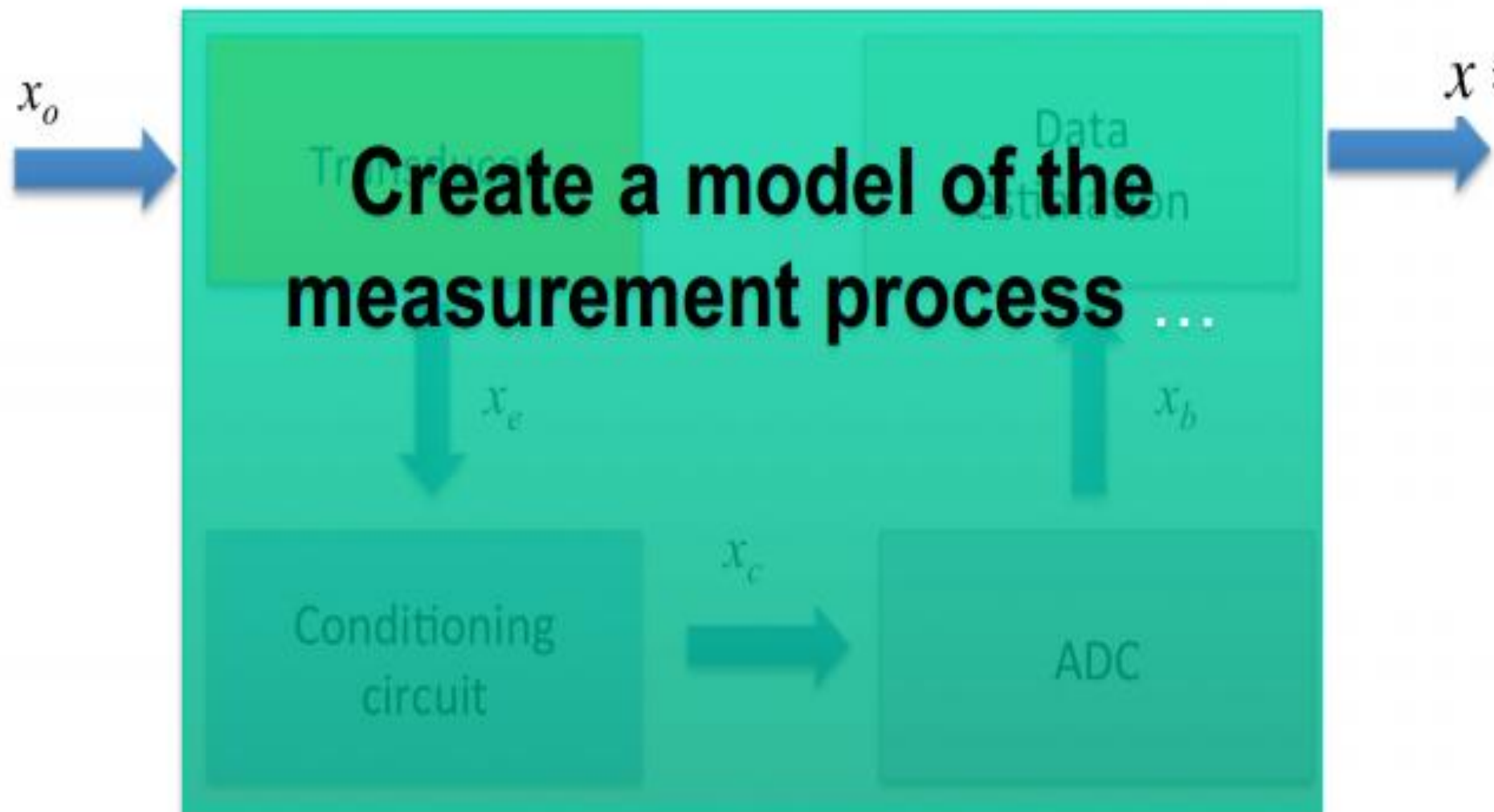
Now it's time for a quick brainstorming...

Please, try to answer the following question:

*how can all these uncertainties in the measurement process affect the application/theory/technique I'm studying or considering?*



## Abstracting the measurement chain ...





## Modeling the measurement process: the additive or “signal plus noise” model

- The whole measurement process can now be seen as a black box, suitably described by an input-output model whose simplest, but generally effective form, is

$$x = x_o + \eta$$

Considered  
Model

where

$$\eta = f_{\eta}(0, \sigma_{\eta}^2) \quad \text{independent and identically distributed (i.i.d.) random variable}$$

The model implicitly assumes that the noise does not depend on the working point  $x_o$

## Modeling the measurement process: the multiplicative model

- Another common model for the sensor is the multiplicative one where

$$x = x_o + \eta x_o = x_o(1 + \eta)$$

In this way, the noise depends on the working point  $x_o$ .  
In absolute terms, the impact of the noise on the signal is  $x_o\eta$ ,  
but the relative contribution is  $\eta$  and does not depend on  $x_o$



Now we have all the theoretical tools to define

- 1) Accuracy
- 2) Precision
- 3) Resolution

These are crucial properties to  
assess the quality of a  
measurement process



## 1) Accuracy

- We say that a measure is accurate when the expectation taken w.r.t. the noise satisfies

$$E[x] = x_o$$

- In order to have an accurate measurement, the instrument and the measurement process have not to introduce any bias contribution

*bias term*

$$x = x_o + k + \eta$$

$$E[x] = x_o + k$$



## Accuracy: how to remove the bias?

- **When the measurement process is biased** we need to subtract the expected value (or its estimate) from the read value.
- However, since  $k$  is unknown, we must rely on a reference value to **estimate it**
- E.g., if we are able to drive the sensor to a controlled state where the expected value is known, say  $x_o$ , then

$$k = E[x] - x_o$$

SENSOR CALIBRATION





## Example: Sensor Calibration

- We bought a **low cost temperature sensor** and are **not sure about its accuracy**. We wish to quantify the potential bias value so as to zero center subsequent measurements.
- To this purpose, **we drive our sensor to operate at a known reference value**  $x_o$  and wait until the dynamics effect associated with the change of state vanishes.
- In the steady state the sensor shares the same temperature as the environment:

$$\hat{k} = \frac{1}{n} \sum_{i=1}^n x_i - x_o$$

- We iterate the process for different  $x_o$  values to **get a calibration curve**



## 2) Precision

- Under the signal plus noise framework, each taken measurement is seen as **a realization of a random variable**.
- Measurements will then be spread around a given value ( $x_0$  in the case of accurate sensors,  $x_0 + k$  in case of an inaccurate one), with the **standard deviation** defining a scattering level index
- **Precision is a measure of such scattering and is function of the standard deviation of the noise**

Given a confidence level  $\delta$ , precision defines an interval  $I$  for  $x_0$  within which all values are indistinguishable due to the presence of uncertainty  $\eta$ : all values  $x \in I$  are equivalent estimates of  $x_0$



## An example of precision: Gaussian Distribution

- Gaussian distribution  $f_{\eta}(0, \sigma_{\eta}^2)$ . *The Gaussian hypothesis holds in many off-the-shelf integrated sensors*
- Under the Gaussian assumption and a confidence level  $\delta$ , we have that a realization  $x_i$  of  $x_o$  lies in
  - $I = [x_o - 2\sigma_{\eta}, x_o + 2\sigma_{\eta}]$  with probability 0.95
  - $I = [x_o - 3\sigma_{\eta}, x_o + 3\sigma_{\eta}]$  with probability 0.997
- The interval defines the precision (interval) of the measure at a given confidence  $\delta$ .
- For example, the precision of the sensor (**sensor tolerance**) could be defined as  $3\sigma_{\eta}$ , so that  $x = x_o \pm 3\sigma_{\eta}$





## Precision: Unknown Distribution

- When  $f_\eta$  is unknown, we cannot use the strong results valid for the Gaussian distribution
- In this case, we need to define an interval / function of  $\delta$  within a **pdf-free framework**
- The issue can be solved by invoking the **Tchebychev theorem** which, given a positive  $\lambda$  value and a confidence  $\delta$ , grants the inequality

$$\Pr(|x_o - x| \leq \lambda \sigma_\eta) \geq 1 - \frac{1}{\lambda^2} = \delta$$

- By selecting a wished confidence  $\delta$ , e.g.,  $\delta = 0.95$ , we select the consequent value  $\lambda$
- The precision interval / is now  $x = x_o \pm \lambda \sigma_\eta$



## Precision: Unknown Distribution (2)

- The lack of priors about the distribution is a cost we pay in terms of a larger tolerance interval

| Distribution      | $\lambda = 1$ | $\lambda = 2$ | $\lambda = 3$ | $\lambda = 4$ |
|-------------------|---------------|---------------|---------------|---------------|
| Gaussian          | 0.682         | 0.954         | 0.997         | 1             |
| Distribution-free | n.a.          | 0.750         | 0.889         | 0.938         |

*By having a priori information about the noise distribution, the precision interval can be easily characterized with a better precision*



### 3) Resolution

- Whereas precision is a property associated with a measure, **resolution** is associated with an instrument/sensor and **represents the smallest value that can be perceived and differentiated by others given a confidence level**
- **Example:** if our instrument has a resolution of 1g, we will not be able to measure values of 1mg due to the limits of the instrument: the scale will make sense in steps of 1g (and all values in such interval will be equivalent and indistinguishable)
- But ...



## Question

**Does a high resolution sensor imply high precision or accuracy?**

**No, having a high resolution neither implies that the measure is accurate nor precise**





## A real sensor example: a temperature sensor for aquatic monitoring



| Features      | Value                                |
|---------------|--------------------------------------|
| Range         | $-4^{\circ}$ to $36^{\circ}\text{C}$ |
| Resolution    | $0.01^{\circ}\text{C}$               |
| Accuracy      | $\pm 0.3^{\circ}\text{C}$            |
| Response time | $\leq 2\text{s}$                     |

- The resolution of the instrument is high, but the impact of the noise on the readout value is high as well.
- The sensor provides values within a  $[-4^{\circ}\text{C}, 36^{\circ}\text{C}]$  interval with an additive error model influencing up to  $\pm 0.3$ .
- We immediately derive that  $\sigma_{\eta} = 0.1$  since the sensor is ruled by a Gaussian distribution from data-sheet information and we consider  $\lambda = 3$



# Questions

